PROJECT REPORT

TASK 28: OUTLIER DETECTION USING PYTHON

1. Abstract

This project focuses on identifying potential outliers in numerical data using Python, Pandas, NumPy, and Matplotlib.

An employee salary dataset containing 30 sample records is analyzed using a box plot and the Interquartile Range (IQR) method.

The project calculates quartiles, the IQR, and outlier boundaries to identify unusually high or low salary values. The results are used to understand the distribution of employee salaries and explain the importance of investigating unusual observations before making data-cleaning decisions.

2. Introduction

Outlier detection is an important part of exploratory data analysis.

An outlier is a numerical observation that differs significantly from the general pattern of the dataset.

Outliers may occur because of data-entry errors, measurement problems, or genuine unusual observations.

A box plot provides a visual representation of numerical data and helps identify values outside the expected range.

The IQR method provides a statistical approach for identifying potential outliers.

This project uses employee monthly salary data to demonstrate these techniques.

3. Objectives

The main objectives are:

- To create a box plot for a numerical variable.
- To understand the distribution of employee salaries.
- To calculate quartiles and the Interquartile Range.
- To determine lower and upper outlier boundaries.
- To identify potential outlier records.

- To understand the importance of investigating unusual values.
- To explain the findings using statistical calculations and visualization.

4. Tools and Technologies

Python: Used to develop and execute the analysis.

Pandas: Used to create, load, and process the employee dataset.

NumPy: Used for numerical operations.

Matplotlib: Used to create the box plot visualization.

Google Colab: Used as the Jupyter Notebook execution environment.

CSV: Used to store the dataset and outlier results.

5. Dataset Description

The project uses a manually created employee salary dataset containing 30 fictional employee records.

The dataset includes three columns:

Column	Description
employee_id	Unique employee identifier
department	Employee department
monthly_salary_inr	Monthly salary in Indian Rupees

The dataset contains one intentionally included unusually high salary to demonstrate potential outlier detection.

The data is created for educational purposes and does not represent actual company employee salaries.

6. Methodology

The project follows these steps:

1. Import Pandas, NumPy, and Matplotlib.
2. Create the employee salary dataset.
3. Export and load the dataset using CSV.
4. Inspect the dataset and its data types.
5. Calculate descriptive statistics.
6. Calculate the first and third quartiles.
7. Calculate the Interquartile Range.

8. Determine the lower and upper outlier boundaries.
9. Create a box plot using Matplotlib.
10. Identify records outside the calculated boundaries.
11. Display potential outlier employee records.
12. Prepare statistical observations.
13. Export the dataset, outlier results, and summary.

7. Understanding Box Plots

A box plot is a graphical representation used to understand the distribution of numerical data.

It contains the following components:

Minimum and Lower Whisker: Represents the lowest non-outlier observation.

First Quartile (Q1): Represents the 25th percentile.

Median (Q2): Represents the middle value of the dataset.

Third Quartile (Q3): Represents the 75th percentile.

Upper Whisker: Represents the highest non-outlier observation.

Outlier Points: Represent observations outside the whisker boundaries.

In this project, the box plot is created using Matplotlib's `plt.boxplot()` function.

8. Interquartile Range Method

The Interquartile Range is calculated using:

IQR = Q3 - Q1

The lower outlier boundary is calculated using:

Lower Boundary = Q1 - 1.5 × IQR

The upper outlier boundary is calculated using:

Upper Boundary = Q3 + 1.5 × IQR

Values below the lower boundary or above the upper boundary are identified as potential outliers.

9. Statistical Calculations

The analysis produces the following results:

Measurement	Value
Total employees	30
Minimum salary	₹35,000
Maximum salary	₹3,50,000
First quartile (Q1)	₹48,500
Median (Q2)	₹61,000
Third quartile (Q3)	₹79,500
Interquartile Range	₹31,000
Lower boundary	₹2,000
Upper boundary	₹1,26,000

The results show that most employee salary observations are concentrated within the central quartile range, while one value is significantly higher than the upper boundary.

10. Box Plot Visualization

A horizontal box plot is created using Matplotlib's `plt.boxplot()` function.

The visualization represents the employee salary distribution and highlights the unusually high salary outside the upper whisker.

Insert the actual employee_salary_boxplot.png image here.

11. Identified Potential Outliers

The IQR method identifies the following record:

Employee ID	Department	Monthly Salary
EMP030	Management	₹3,50,000

The identified salary of ₹3,50,000 exceeds the calculated upper boundary of ₹1,26,000.

No observations fall below the lower boundary.

Therefore, the dataset contains one potential high outlier and zero lower outliers.

12. Key Observations

The analysis provides the following observations:

1) The dataset contains 30 employee salary records.
2) The middle 50% of salaries lies between ₹48,500 and ₹79,500.

3) The upper outlier boundary is ₹1,26,000.
4) One employee record has a salary of ₹3,50,000, exceeding the upper boundary.
5) No salaries fall below the lower boundary.
6) The identified value is retained in the original dataset for further investigation.

13. Importance of Outlier Detection

Outlier detection helps analysts understand unusual observations in numerical datasets.

Unusual values can influence averages, statistical calculations, and machine-learning models.

Identifying these values helps determine whether they represent genuine observations or data-quality problems.

Outliers should not automatically be removed because they may contain important information.

14. Applications

Outlier detection is useful in:

- Employee salary analysis.
- Banking and financial transactions.
- Fraud detection.
- Housing price analysis.
- Sales data analysis.
- Healthcare data analysis.
- Machine-learning data preparation.

15. Limitations

The project uses a small, manually created dataset containing 30 employee records.

The IQR method identifies statistically unusual observations but cannot determine whether an observation is incorrect.

A high salary may be legitimate depending on an employee's role, experience, and responsibilities.

Additional contextual information is required before deciding whether any value should be modified or removed.

16. Future Enhancements

The project could be extended by:

- Using larger real-world employee datasets.
- Comparing multiple outlier detection techniques.
- Applying Z-score analysis.
- Creating box plots for several numerical variables.
- Investigating unusual salary records using employee roles.
- Developing automated data-quality reporting.

17. Conclusion

This project demonstrates how Python and Matplotlib can be used to visualize numerical data through box plots and identify potential outliers using the IQR method.

The analysis identifies one unusually high salary of ₹3,50,000 that exceeds the upper boundary of ₹1,26,000.

The identified value is retained in the original dataset because an unusual observation is not necessarily an error.

This exercise provides practical knowledge of statistical outlier detection and data visualization.

18. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. NumPy Documentation: https://numpy.org/doc/
3. Matplotlib Documentation: https://matplotlib.org/stable/
4. Matplotlib Box Plot Reference: https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.boxplot.html

20. Project Links

GitHub Repository: https://github.com/venky1318/Outlier_Detection